数据库

理论部分

数据库概念

1. **数据库**：数据存储的仓库,有组织,规范的存储数据
2. 数据库解决问题
    - 数据的长久储存
    - 数据规范化储存管理从而高效查找数据
    - 数据备份储存,储存重要数据

sql基本语法

1. ==通用: 分号;结尾,--或#单行注释,/*多行注释*/==

2. **分类**

    - DDL定义数据库对象(数据库,表)
    - DML数据操作,增删改
    - DQL数据查询
    - DCL增加数据库用户,改变权限

3. **DDL**

    1. 数据库操作
    - show databases;查询所有
    - select database();查询当前数据库
    - create database [库名];创建数据库
    - drop database [库名];删除数据库
    - use [库名];使用,切换数据库
    2. 表操作
    - show tables;
    - ==desc [表名];查询表结构==
    - show create table [表名]查询指定表的建表语句
    - ==create table [表名](数据名 类型 comment '注释',...)comment '注释';== 在当前数据库创建
    - alter table [表名] add 数据名 类型 comment '注释';增
    - alter table [表名] modify 数据名 新数据类型;修改数据类型
    - alter table [表名] change 数据名 新数据名 类型;修改数据名
    - alter table [表名] drop 数据名;删除
    - alter table 表名 rename to 新表名

4. **DML**

    - insert into [表名] (数据名) values (值)
    - update [表名] set 数据名=值1,...[where条件];
    - delete from [表名] [where条件]

5. **DQL**
    - ==select 数据名,... from 表名;通配符*查询所有字段==
    - select 数据名[as 别名],... from 表名;
    - select distinct 数据列表 from 表名;去除重复记录

实践部分

1. **安装mysql**
    1. Ubuntu打开终端
    2. sudo apt update
    3. sudo apt install mysql-server

```
wind@wind-VMware-Virtual-Platform:~$ sudo apt update
命中:1 http://mirrors.tuna.tsinghua.edu.cn/ubuntu noble InRelease
命中:2 http://mirrors.tuna.tsinghua.edu.cn/ubuntu noble-updates InRelease
命中:3 http://mirrors.tuna.tsinghua.edu.cn/ubuntu noble-backports InRelease
命中:4 http://security.ubuntu.com/ubuntu noble-security InRelease
正在读取软件包列表... 完成
正在分析软件包的依赖关系树... 完成
正在读取状态信息... 完成
有 3 个软件包可以升级。请执行 'apt list --upgradable' 来查看它们。
wind@wind-VMware-Virtual-Platform:~$ sudo apt install mysql-server
正在读取软件包列表... 完成
正在分析软件包的依赖关系树... 完成
正在读取状态信息... 完成
将会同时安装下列软件:
  libaio1t64 libcgi-fast-perl libcgi-pm-perl libevent-core-2.1-7t64
  libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64
  libhtml-template-perl libmecab2 libprotobuf-lite32t64 mecab-ipadic
  mecab-ipadic-utf8 mecab-utils mysql-client-8.0 mysql-client-core-8.0
  mysql-common mysql-server-8.0 mysql-server-core-8.0
建议安装:
  libipc-sharedcache-perl mailx tinyca
下列【新】软件包将被安装:
  libaio1t64 libcgi-fast-perl libcgi-pm-perl libevent-core-2.1-7t64
  libevent-pthreads-2.1-7t64 libfcgi-bin libfcgi-perl libfcgi0t64
```

    4. sudo mysql_secure_installation

5. y 设置密码 y y y y

```
wind@wind-VMware-Virtual-Platform:~$ sudo mysql_secure_installation

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.

VALIDATE PASSWORD COMPONENT can be used to test passwords
and improve security. It checks the strength of password
and allows the users to set only those passwords which are
secure enough. Would you like to setup VALIDATE PASSWORD component?

Press y|Y for Yes, any other key for No: y

There are three levels of password validation policy:

LOW    Length >= 8
MEDIUM Length >= 8, numeric, mixed case, and special characters
STRONG Length >= 8, numeric, mixed case, special characters and dictionary
            file
```

6. 检查systemctl status mysql.service

```
wind@wind-VMware-Virtual-Platform:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
     Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: en
     Active: active (running) since Sat 2025-09-13 19:53:54 CST; 3min 59s ago
    Process: 13311 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=
   Main PID: 13320 (mysqld)
     Status: "Server is operational"
      Tasks: 38 (limit: 4545)
     Memory: 361.1M (peak: 378.0M)
        CPU: 2.205s
     CGroup: /system.slice/mysql.service
             └─13320 /usr/sbin/mysqld

9月 13 19:53:53 wind-VMware-Virtual-Platform systemd[1]: Starting mysql.service
9月 13 19:53:54 wind-VMware-Virtual-Platform systemd[1]: Started mysql.service
```

7. sudo mysql -u root -p出现mysql>则成功

```
wind@wind-VMware-Virtual-Platform:~$ sudo mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 8.0.43-0ubuntu0.24.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

2. **表**

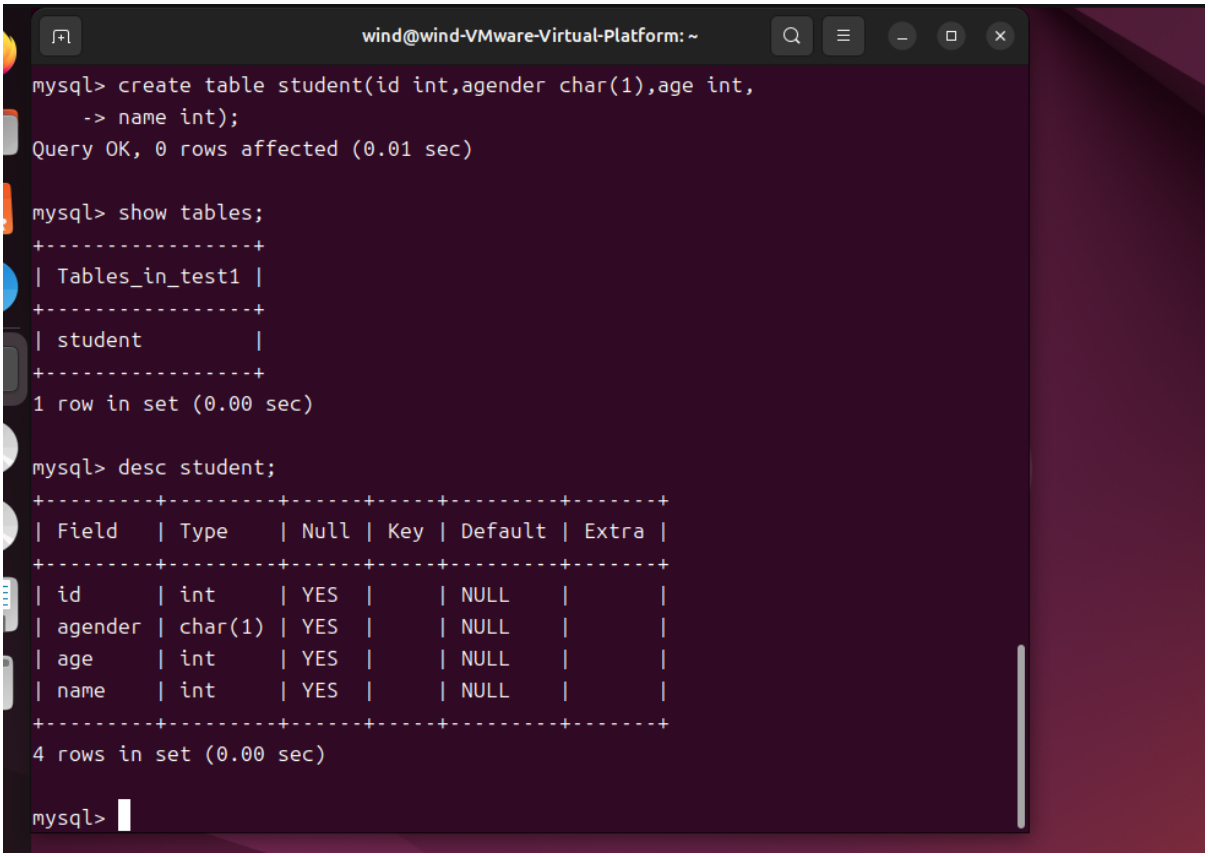
1. 数据库

```
mysql> create database test1;
Query OK, 1 row affected (0.01 sec)

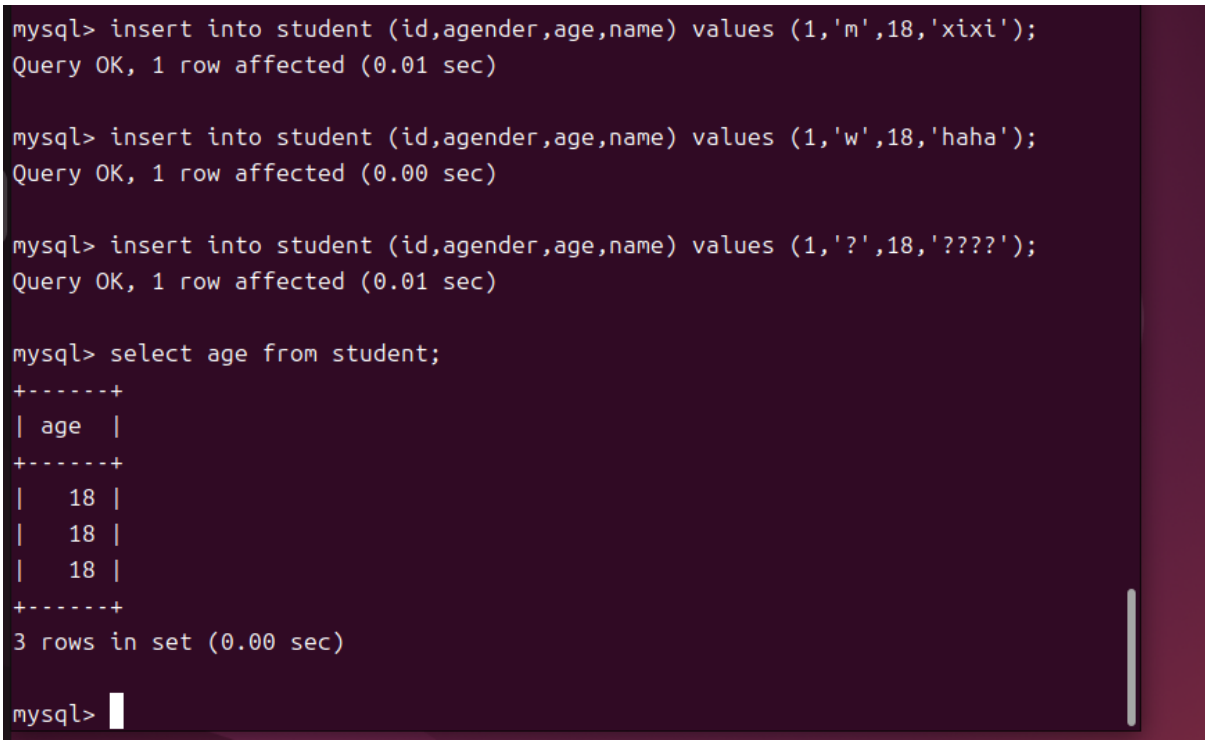
mysql> show databases
    -> ;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
| test1              |
+--------------------+
5 rows in set (0.00 sec)

mysql> use database test1
ERROR 1049 (42000): Unknown database 'database'
mysql> use test1;
Database changed
```

2. 表


wind@wind-VMware-Virtual-Platform: ~
mysql> create table student(id int,agender char(1),age int,
-> name int);
Query OK, 0 rows affected (0.01 sec)
mysql> show tables;
+----------------+
| Tables_in_test1 |
+----------------+
| student |
+----------------+
1 row in set (0.00 sec)
mysql> desc student;
+---------+---------+------+-----+---------+-------+
| Field | Type | Null | Key | Default | Extra |
+---------+---------+------+-----+---------+-------+
id	int	YES		NULL	
agender	char(1)	YES		NULL	
age	int	YES		NULL	
name	int	YES		NULL	
+---------+---------+------+-----+---------+-------+
4 rows in set (0.00 sec)
mysql>

3. 插入


mysql> insert into student (id,agender,age,name) values (1,'m',18,'xixi');
Query OK, 1 row affected (0.01 sec)
mysql> insert into student (id,agender,age,name) values (1,'w',18,'haha');
Query OK, 1 row affected (0.00 sec)
mysql> insert into student (id,agender,age,name) values (1,'?',18,'????');
Query OK, 1 row affected (0.01 sec)
mysql> select age from student;
+------+
| age |
+------+
| 18 |
| 18 |
| 18 |
+------+
3 rows in set (0.00 sec)
mysql>

4. 查询

```
mysql> select * from student;
+------+---------+------+------+
| id   | agender | age  | name |
+------+---------+------+------+
|    1 | m       |   18 | xixi |
|    1 | w       |   18 | haha |
|    1 | ?       |   18 | ???? |
+------+---------+------+------+
3 rows in set (0.00 sec)

mysql>
```

3. **navicat**

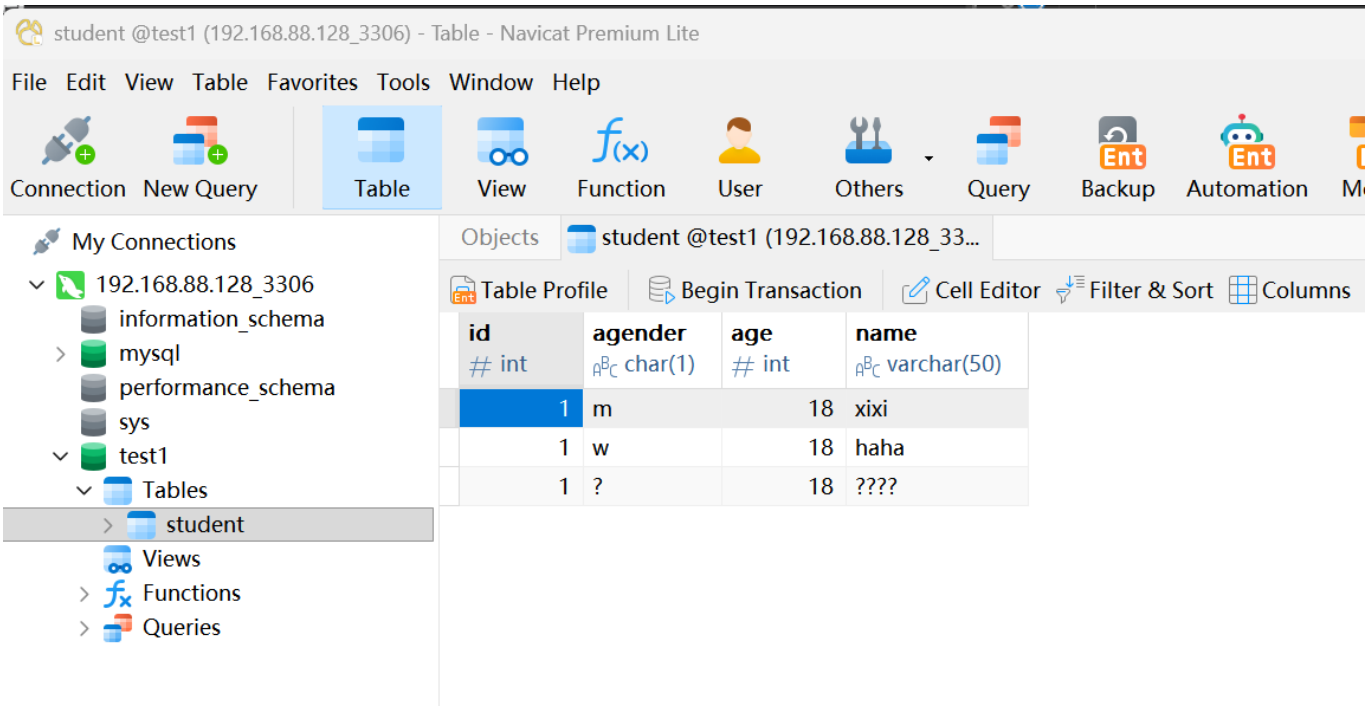
1. **添加新用户**

- sql查看密码策略show variables like 'validate_password%';
- create user 'your_username'@'你的主机ip' identified '密码';==密码符合策略==
- grant all privileges on . 'your_username'@'密码' with grant option;
- flush privileges;刷新

2. **或修改root权限**

- update mysql.user set host='%' where user='root';
- 刷新

3. sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf**修改mysql配置**bind-address=0.0.0.0后重启



加分项

1. 关系型数据库:建立在关系模型基础上,由多张二维表相互连接组成
2. 常见数据库类型(统称Nosql)
    - 文档数据库
    - 图数据库
    - 列族数据库
    - ==键值数据库(Redis)==
3. 差别
    - 结构不同二维表,文档,图...
    - 语言,关系型用sql,Nosql型没统一语言
    - 表结构严格定义,nosql类型每个结构可以不同
    - 运用场景不同